

ESP32 MQTT Home Assistant DIY Kit

Installation and configuration guide for MQTT and ESPHome firmware, including ESP32-C6 Super Mini

About this guide

The kit can run either the PlatformIO firmware with MQTT and its own web dashboard, or the alternative ESPHome firmware with direct Native API integration. The ESPHome section provides the complete ESP32-C6 Super Mini procedure.

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The model is usually printed on the metal module or on the back of the development board. The firmware profile must match the supplied board.

ESP32-WROOM-32 / ESP32 Dev Module

Dual-core Xtensa LX6 processor with 2.4 GHz Wi-Fi, Bluetooth Classic and Bluetooth LE. It is the mature general-purpose profile for sensors and automation.

Onboard LED: depends on the DevKit board; the firmware defaults to an external heartbeat LED on GPIO18.

ESP32-C3-DevKitM-1

Single-core RISC-V processor with Wi-Fi and Bluetooth LE 5. Suitable for compact sensors and automation nodes.

Onboard RGB LED: GPIO8.

ESP32-C6-DevKitC-1

RISC-V processor with Wi-Fi 6, Bluetooth LE 5 and an IEEE 802.15.4 radio.

Onboard RGB LED: GPIO8.

ESP32-C6 Super Mini

Compact ESP32-C6 board with USB-C and an integrated USB Serial/JTAG interface. It is supported by the MQTT, Zigbee Router and ESPHome profiles.

USB: GPIO12/GPIO13 are reserved.

BOOT: GPIO9. **RGB:** GPIO8.

ESP32-S3-DevKitC-1

Dual-core Xtensa LX7 processor with native USB and vector instructions for AI/DSP workloads.

Onboard RGB LED: GPIO48 on the initial revision or GPIO38 on v1.1. The dashboard reports the pin used by the firmware.

Available hardware capabilities not enabled by this firmware

ESP32-C6 can run Zigbee/Thread, while C6 and S3 can be used for Matter. The MQTT firmware, Zigbee Router firmware and ESPHome firmware are separate images and cannot run simultaneously on the same board.

E ESPHome on ESP32-C6 Super Mini

The `c6-supermini` profile uses ESP-IDF and integrates the DHT11, PIR sensor, relay and OLED directly with Home Assistant through the ESPHome Native API. An MQTT broker is not required.

Function	Default GPIO
DHT11	GPIO0
PIR	GPIO1
Active-LOW relay	GPIO2
External heartbeat LED	GPIO3
OLED SDA / SCL	GPIO20 / GPIO19

Linux installation

- 1 Connect the board directly with a USB-C data cable. It normally appears as `/dev/ttyACM0`.
- 2 From the project directory, run `cd esphome`, followed by `./setup.sh init`.
- 3 Validate the profile with `./setup.sh check c6-supermini`.
- 4 Compile and flash it with `./setup.sh run c6-supermini /dev/ttyACM0`.

Windows installation

- 1 Install Python 3.11 or newer and find the board port in Device Manager, for example `COM5`.
- 2 Open PowerShell in the `esphome` directory and run `.\setup.ps1 init`.
- 3 Validate the profile with `.\setup.ps1 check c6-supermini`.

- 4 Compile and flash it with `.\setup.ps1 run c6-supermini COM5` . From Command Prompt, use `setup.cmd run c6-supermini COM5` .

If flashing does not start

Hold BOOT, press and release RESET, release BOOT, then immediately start the upload command. Press RESET after flashing. Do not attach peripherals to GPIO12 or GPIO13 because they are the USB D-/D+ lines.

Initial Wi-Fi setup

- 1 After boot, connect to the `HA Kit C6 Super Mini` access point using the `fallback_password` value from `esphome/secrets.yaml` .
- 2 Open `http://192.168.4.1` , select the customer's 2.4 GHz Wi-Fi network and enter its password.
- 3 After the board joins the router, it is available at `esp32-ha-kit-c6-supermini.local` or its DHCP address.

Add the device to Home Assistant

- 1 Open **Settings** → **Devices & services** → **Add integration** → **ESPHome**.
- 2 Enter `esp32-ha-kit-c6-supermini.local` or the IP address.
- 3 When prompted for the encryption key, enter `api_encryption_key` from `secrets.yaml` .
- 4 The local web page separately uses `web_username` and `web_password` . The OTA password is stored as `ota_password` .

Customer handover

The captive portal configures Wi-Fi only. The API key, OTA password and web authentication are compiled into the firmware. For exclusive control, the customer must adopt or rebuild the board in their own ESPHome Device Builder with their own credentials.

2 Installation requirements

- ✓ A router providing a **2.4 GHz Wi-Fi** network.
- ✓ A phone, tablet or computer with a web browser.
- ✓ A 5 V USB power source rated for at least 500 mA and a compatible power/data cable.
- ✓ A working Home Assistant installation if MQTT integration is required.
- ✓ Mosquitto Broker and the MQTT integration enabled in Home Assistant.

The USB connector may differ

Depending on the board and revision, the connector can be USB-C or Micro-USB. WROOM boards commonly use a CP210x or CH340 USB-to-UART bridge, while some S3 boards provide separate USB-to-UART and native USB ports.

3 First startup

- 1 Power the board through the appropriate USB port.
- 2 Wait approximately 10-15 seconds for firmware initialization.
- 3 If no Wi-Fi credentials are stored, the board creates the `ESP32_HAKit` network.
- 4 Connect your phone or computer using password `12345678`.
- 5 Open `http://192.168.4.1` in a browser.

Access Point mode

SSID: `ESP32_HAKit` Password: `12345678` Address: `192.168.4.1`

4 Configure Wi-Fi and find the IP address

- 1 Open the **WiFi** tab in the ESP32 dashboard.
- 2 Enter the home network name and password.
- 3 Press **Save and connect** (**Salveaza si conecteaza**).
- 4 Reconnect your phone or computer to the home network.
- 5 Open the dashboard using one of the methods below.

Method	Address or action
Local hostname	<code>http://esp32-ha-kit.local</code>
Router	Find the <code>esp32-ha-kit</code> client in the DHCP list.
OLED	The IP is displayed when an OLED is connected and enabled.
Serial log	At 115200 baud, look for the <code>IP Address:</code> line.

Unsupported network

These boards use 2.4 GHz Wi-Fi. If a combined 2.4/5 GHz SSID causes connection problems, temporarily create a separate 2.4 GHz network name.

5 Configure MQTT

- 1 Open the board dashboard and select the **MQTT** tab.
- 2 Enter the broker address, port and authentication details.
- 3 Use a unique Client ID when installing multiple kits.
- 4 Press **Save and apply** (`Salveaza si aplica`).

Field	Example
Broker	<code>192.168.1.50</code> - Home Assistant/Mosquitto IP
Port	<code>1883</code>
Client ID	<code>esp32-ha-kit</code> or a unique name
Username / password	Credentials accepted by the MQTT broker

The status bar turns green and displays
MQTT: Connected
after successful authentication.

6 Configure components and pins

Open the **Hardware** tab. Select the correct pin for each installed component. Set components that are not connected to **Disabled / not configured** (**Dezaktivat / neconfigurat**).

Profile	DHT	PIR	Relay	Heartbeat	SDA	SCL	Onboard RGB
ESP32-WROOM-32	4	32	23	18	21	22	—
ESP32-C3	4	3	7	10	6	5	8
ESP32-C6	2	3	7	18	6	10	8
ESP32-C6 Super Mini	0	1	2	3	20	19	8*
ESP32-S3	4	5	7	18	8	9	48/38*

ESP32-WROOM-32 does not guarantee an addressable onboard RGB LED. * The S3 RGB pin depends on the board revision; on C6 Super Mini, GPIO8 is reserved for the onboard RGB LED and is not the ESPHome heartbeat LED. Do not manually assign reserved pins.

How are components detected?

DHT11 is shown after valid readings and the OLED after a successful I2C response. PIR sensors and relays cannot be reliably identified electrically; they are shown when enabled in the Hardware tab.

Power levels

Use 3.3 V for the DHT11 and OLED. Many HC-SR501 and relay modules use a 5 V supply, but signals connected to the ESP32 must remain compatible with 3.3 V logic. Always check the supplied module.

Tab	Function
Dashboard	Shows detected or enabled components only; controls the relay and onboard RGB LED.
MQTT	Broker settings and connection status.
WiFi	Change or delete the stored network.
Hardware	Enable/disable components, select GPIOs and inspect the board pinout.
Serial Log	A copy of the board's internal log, available directly in the browser.
Board Info (Info Placa)	Model, firmware profile, memory, MAC address, uptime and capabilities.

Interface language

Use the selector at the top of the page to choose Romanian or English. The selection is stored in the board's NVS memory and applies to every client opening the dashboard.

Onboard RGB LED

Color, brightness and ON/OFF controls are available without external wiring on profiles that declare an RGB LED. The WROOM profile does not show this card unless the board declares a compatible RGB LED.

- 1 In Home Assistant, open **Settings** → **Devices & services** → **MQTT**.
- 2 Find the **ESP32 HA Kit** device.
- 3 Entities are published for peripherals enabled in the hardware configuration.
- 4 Add the entities to a dashboard and create the required automations.

Enabled peripheral	Home Assistant entity
DHT11	Temperature and humidity
PIR	Motion binary sensor
Relay	ON/OFF switch

The onboard RGB LED is currently controlled from the ESP32 dashboard and is not published as an MQTT entity.

The ESP32_HAKit network is missing

The board may already have a stored network. Check the router for `esp32-ha-kit` or clear credentials from the WiFi tab.

192.168.4.1 does not open

Connect to `ESP32_HAKit` and temporarily disable mobile data or VPN.

MQTT remains disconnected

Check the broker IP, port 1883, credentials, firewall and that both devices are on the same local network.

A dashboard card is missing

Check the Hardware tab. DHT and OLED must respond; PIR and relay must be explicitly enabled.

The WROOM serial port is missing

Use a USB data cable and install the driver for the board's USB-to-UART bridge, usually CP210x or CH340.

The S3 serial port is missing

Use a USB data cable and the correct port. Depending on the connection, the board may appear through native USB or a USB-to-UART bridge.

C6 Super Mini does not flash

Use a USB-C data cable and the BOOT + RESET sequence described in the ESPHome section. On Linux check `/dev/ttyACM*`; on Windows check Device Manager for the COM port.

ESPHome is not discovered by Home Assistant

Verify that the board and Home Assistant are on the same network, try the IP address instead of the `.local` name, and enter the exact `api_encryption_key` value.

Separate reset actions

Deleting data from the WiFi tab removes Wi-Fi credentials only. MQTT and GPIO settings are managed separately in the MQTT and Hardware tabs.